DS 4002 Case Study Hook Document by Viv Hughes

Exploring Factors Affecting Fertility Rates Worldwide

Modern Homo sapiens, our species, evolved in Africa about 300,000 years ago. In the grand scheme of things, we as humans have been on this planet for a small fraction of time. However, we have clearly taken over and are the dominant species with a population of about 8 billion. So how did we get here? And, arguably more important, where is the world population headed next?

For the purpose of this study you are a researcher working with the UN to improve their future fertility rate predictions worldwide. In this case study you will follow 15 countries: 2 randomly selected from each continent (not including Antarctica and using Oceania instead of just Australia), the United States, and Nigeria and South Korea as they currently have the lowest and highest birth rates. For the countries you will follow the fertility rate (the number of live births per woman) and 5 social factors: GDP per capita, women's educational attainment in years, life expectancy in years, child/infant mortality rate per 100 live births, and the homicide rate per 100,000 people. The data sets all go back at least 40 years. Some countries are missing from certain datasets, but that is alright as no country is missing from a majority of the datasets and each country is solely acting as an indicator to detect trends.

Your first goal is to use a regression analysis (two way fixed effects, TWFE) to see which social factor is the best indicator of fertility rates. Then, using ARIMAX, to combine all of the social factor and fertility rate data for each country to build a predictive model. Once your ARIMAX predictive model is complete you will compare it to the given UN predictions data and report your RMSE and MAE errors.

Everything you need is located in GitHub: example code for TWFE and ARIMAX, previous analysis code, and the datasets. Good luck! The UN is counting on you.